

- 7 A rock of mass $2m$ in deep space, initially travelling at velocity v , explodes into two parts of equal mass, one of which becomes stationary immediately after the explosion.

What is the kinetic energy of the moving part after the explosion?

- A $\frac{1}{2}mv^2$ B mv^2 C $\frac{3}{2}mv^2$ D $2mv^2$

By COM, momentum after explosion = $2mv$, which is only carried by moving part.

So KE of moving part = $(2mv)^2/(2m) = 2mv^2$

Answer: D

2 A single horizontal force F is applied to a block X which is in contact with a concrete block Y , as